

HURMAT FATIMA

Lahore, Pakistan

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Education

Punjab University College of Information Technology – PUCIT

Sep. 2024 – May 2028

Bachelor of Science in Information Technology

Lahore, Pakistan

Relevant Coursework

- Data Structures
- Object Oriented Programming
- Databases
- Operating Systems
- Information Security
- Computer Organization
- Assembly Language
- Discrete Structures
- Computer Networking
- Digital Logic Design
- Artificial Intelligence
- Machine Learning
- Web Technologies

Work Experience

Allied Bank Limited (ABL)

2026

AI Engineering Intern

Lahore, Pakistan

- Built a RAG-based chatbot using FastAPI, ChromaDB, and Ollama, implementing query rewriting, conversation memory, chunk retrieval, and retrieval-quality improvements.
- Designed n8n workflows to orchestrate chatbot components and connected automation nodes to create end-to-end AI workflows.
- Explored Docker and MCP (Model Context Protocol) servers for containerization and AI tool integration.
- Developed a SOC comparison pipeline to extract bank charges from competitor PDFs and compare them against ABL's master taxonomy using LLM-assisted extraction.

Projects

SOC Comparison Pipeline | *Python, Gemini, LLMs, PDF Processing, Excel*

2026

- Built an AI-assisted pipeline to compare Schedule of Charges (SOC) across banks against ABL's master taxonomy.
- Extracted and validated charges from competitor bank PDFs using LLM-assisted analysis and generated structured comparison spreadsheets.

RAG-Based ABL Chatbot | *FastAPI, ChromaDB, Ollama, Python*

2026

- Built a RAG-based chatbot using FastAPI, ChromaDB, and Ollama for document-based question answering and semantic retrieval.
- Implemented query rewriting, conversation memory, chunk retrieval, vague-query detection, and retrieval-quality testing.

NutriAI | *React, Node.js, Express, MongoDB, Groq AI, Nutritionix API*

2026

- Built a full-stack AI nutritionist application with meal planning, BMI calculator, food database search, recipe suggestions, and progress dashboard with charts.
- Integrated Groq AI for personalized weekly meal plans and Nutritionix API for real-time nutritional data; implemented JWT authentication and appointment booking.

ZestMart – General Grocery Store | *React, Node.js, Express, MongoDB*

2026

- Developed a full-stack e-commerce grocery platform with product listings, category filters, shopping cart, order management, and admin inventory dashboard.
- Implemented JWT authentication, payment flow integration, and a responsive React interface with custom CSS.

Encryption & Decryption App | *HTML5, CSS3, JavaScript, Web Crypto API*

2026

- Built a browser-based cryptography application supporting AES-256, RSA key generation, SHA hashing, Base64, Hex, Morse, and XOR image encryption.
- Implemented PBKDF2 key stretching, AES-GCM encryption, RSA-OAEP, and hash integrity verification using the Web Crypto API.

Online Chat Application | *React, Express, Socket.IO, MongoDB*

2025

- Built a real-time chat application enabling private messaging using WebSockets with Socket.IO and Express APIs.
- Implemented login/logout, message persistence in MongoDB, and responsive UI with React.

Online Bookstore Platform | *React, Express, MongoDB*

2025

- Developed a full-stack e-commerce web application for browsing and purchasing books with cart and order management features.
- Implemented JWT authentication and used MongoDB to manage inventory and order data.

Technical Skills

Languages: C++, Python, JavaScript, SQL, PL/SQL, PHP, HTML/CSS
Frameworks & Libraries: React.js, Node.js, Express.js, FastAPI, Bootstrap, Socket.IO, Pandas, NumPy, Matplotlib
Databases: Oracle, MySQL, MongoDB, PostgreSQL
AI & LLM Technologies: Generative AI, RAG, LLMs, Groq AI, Ollama, ChromaDB, Prompt Engineering
APIs & Integration: REST APIs, Nutritionix API, Web Crypto API, MCP
Developer Tools: Git, GitHub, VS Code, Postman, Jupyter Notebook, Docker
Other Technologies: MERN Stack, n8n, JWT Authentication, WebSockets, OOP, File I/O

Awards & Certifications

Full Stack AI Engineer	2024 – 2025
<i>Certified Course Graduate – Arfa Software Technology Park, Lahore</i>	<i>Lahore, Pakistan</i>
- Completed an industry-level course in Full Stack Web Development (MERN Stack) and Artificial Intelligence / Machine Learning. - Gained hands-on experience building full-stack applications with React, Node.js, Express, and MongoDB alongside practical AI/ML model development.	
The Punjab Schools	2019 – 2021
<i>Secured 2nd Position across Lahore Board in Secondary Education (Matriculation, 10th Grade)</i>	<i>Lahore, Pakistan</i>
Apex College	2021 – 2023
<i>Achieved 2nd Rank in College – Higher Secondary Education (F.Sc. Pre-Medical)</i>	<i>Punjab, Pakistan</i>